Conditions 1 Basic Property

Def VIII. 1.1 let $\Sigma$ be a set partially ordered by $\leq$.

① If every increasing sequence $x_1 \leq x_2 \leq \dots$ in $\Sigma$ has, $\exists n>0$. s.t. $x_n = x_{n+1} = \dots$

Then call $\Sigma$ satisfies a.c.c.

② If every decreasing sequence $x_1 \geq x_2 \geq \dots$ in $\Sigma$ $\exists n>0$ s.t. $x_n = x_{n+1} = \dots$

Then call $\Sigma$ satisfies d.c.c.

Def VIII. 1.2. $\Sigma$ is the set of submodules of a module $M$ partially ordered by $\subseteq$.

If $M$ satisfies a.c.c. then call $M$ is a Noetherian module

If $M$ satisfies d.c.c. then call $M$ is an Artinian module

Prop VIII. 1.3. $\Sigma$ satisfies a.c.c $\Leftrightarrow$ Every $P \subseteq \Sigma$ $(P \neq \phi)$ has a maximal element.

pf: (skipped)

Prop. VIII. 1.4. (1) $M$ is a Noetherian module $\Leftrightarrow$ every submodule of $M$ is finitely generated.

In particular, $M$ is a Noetherian ring $\Leftrightarrow$ every ideal is finitely generated.

pf: (skipped).

Prop. VIII. 1.5. Let $0 \longrightarrow M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \longrightarrow 0$. be exact. Then

i). $M$ is Noetherian $\Leftrightarrow$ $M'$ and $M''$ Noetherian.

ii) $M$ is Artinian $\Leftrightarrow$ $M'$ and $M''$ Artinian.

pf: i). $\Rightarrow$ let $M_1' \subseteq M_2' \subseteq \dots \subseteq M'$ then $\alpha(M_1') \subseteq \alpha(M_2') \subseteq \dots \subseteq M$ is stationary.

$\alpha$ is injective $\Rightarrow$ $\alpha^{-1} \circ \alpha = id$ $\Rightarrow$ $M_1' \subseteq \dots \subseteq M'$ is stationary.

Similare to $M''$

$\Leftarrow$ let $M_1 \subseteq M_2 \subseteq \dots \subseteq M$ then $\alpha^{-1}(M_1) \subseteq \dots \subseteq \alpha^{-1}(M)$ is stationary.

$\beta(M_1) \subseteq \dots \subseteq \beta(M)$ is stationary.

$\exists n>0$. s.t. $\alpha^{-1}(M_n) = \alpha^{-1}(M_{n+1}) = \dots$ ; $\beta(M_n) = \beta(M_{n+1}) = \dots$

Thus $\alpha(\alpha^{-1}(M_n)) = M_n \cap \operatorname{Im}\alpha$

$\beta^{-1}(\beta(M_n)) = M_n + \ker\beta = M_n + \operatorname{Im}\alpha$

$\forall x \in M_{n+1}$ $\exists y \in M_n$. s.t. $\beta(x) = \beta(y) \Rightarrow \begin{cases} x-y \in \operatorname{Im}\alpha \\ x-y \in M_{n+1} \end{cases} \Rightarrow x-y \in M_{n+1} \cap \operatorname{Im}\alpha = M_n \cap \operatorname{Im}\alpha$

$\Rightarrow x = y + (-y+x) \in M_n$ ▨

Corollary VIII 1.6 ① $M_i$ (finitely many) Noetherian (Artinian) $\Rightarrow \bigoplus_{i=1}^{n} M_i$ Noetherian (Artinian)

② $M \subseteq N$ · Noetherian (Artinian) $\Rightarrow$ $N/M$ Noetherian (Artinian)

③ $A$ is a Noetherian (Artinian) ring $\Rightarrow$ Any finitely-generated $A$-module $M$ Noetherian (Artinian)

Def VII 1.7 A composition of $M$ is a maximal chain, $M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq M_n = 0$ where $M_{i0}/M_{i+1}$ has no submodules except 0 and itself.

Prop VIII.1.8. (1) Suppose that $M$ has a composition series of length $n$. Then every composition series of $M$ has length $n$. and every chain in $M$ can be extended to a composition series

(2) $M$ has a composition series $\Leftrightarrow$ $M$ satisfies a.c.c. & d.c.c.

(3) let $0 \to M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \to 0$ exact, then $l(M) = l(M') + l(M'')$.

Pf: (skipped) ▨

Prop VII.1.9. For $k$-vector spaces $V$, TFAE: i) finite dimension; ii) finite length; iii) a.c.c. iv). d.c.c.

Pf: i) $\Leftrightarrow$ ii) obviously. ii) $\Rightarrow$ (iii), ii) $\Rightarrow$ (iv) are from (1.8) iii) $\Rightarrow$ (i) obvious ▨ iv) $\Rightarrow$ (i)

Prop VIII.1.10. $A$ be a ring. $0 = \mathfrak{m}_1 \mathfrak{m}_2 \cdots \mathfrak{m}_n$. $\mathfrak{m}_i$ are maximal. Then $A$ is Artinian $\Leftrightarrow$ Noetherian

Pf: $A \supsetneq \mathfrak{m}_1 \supsetneq \mathfrak{m}_1\mathfrak{m}_2 \supsetneq \cdots \supsetneq \mathfrak{m}_1 \cdots \mathfrak{m}_n = 0$ $\mathfrak{m}_1 \cdots \mathfrak{m}_i / \mathfrak{m}_1 \cdots \mathfrak{m}_{i+1}$ is a vector space over $A/\mathfrak{m}_{i+1}$

$\Rightarrow$: a.c.c $\Leftrightarrow$ d.c.c. (Since let $I_k = \mathfrak{m}_1 \cdots \mathfrak{m}_k$, then $0 \to I_k \to I_{k-1} \to I_{k-1}/I_k \to 0$. exact

▨

## Noetherian Rings

VIII 2.1 (Hilbert Basis Theorem). If $A$ is Noetherian, then $A[x]$ is Noetherian.

Pf: Let $\mathfrak{a}$ be an ideal in $A[x]$.

Then $I := \{$the leading coefficients in $\mathfrak{a}\}$ is an ideal in $A$

Thus $I = (a_1, \dots, a_n)$ $\forall i \in [1,n]$. $\exists f_i \in \mathfrak{a}$ s.t. $f_i = a_i x^{r_i} + \dots$ $\deg f_i = r_i$

Let $r = \max_{1 \le i \le n} r_i$ Then $f_i$ generate $\mathfrak{a}' \subseteq \mathfrak{a}$ in $A[x]$.

Let $f \in \mathfrak{a}$. where $f = a x^m + \dots$, $\deg f = m$ and $a \subseteq I$

If $m \ge r$ let $a = \sum_{i=1}^{n} u_i a_i$ $(a_i \in I,\ u_i \in A)$

Then $f - \sum_{i=1}^{n} u_i f_i x^{m-r_i} \in \mathfrak{a}$. and the degree $\le m-1$

By continuing doing that, we have $f = g + h$, where $\deg g < r$. and $h \in \mathfrak{a}'$

Let $M$ be the $A$-module generated by $1, \dots, x^{r-1}$

And we have $\mathfrak{a} = (\mathfrak{a} \cap M) + \mathfrak{a}'$

$M$ is finitely-generated $A$-module $\Rightarrow$ $M$ is Noetherian

Since $\mathfrak{a} \cap M \subseteq M$ is a submodule $\Rightarrow$ $\mathfrak{a} \cap M$ finitely-generated $:= (g_1, \dots, g_m)$

$\Rightarrow$ $\mathfrak{a}$ is generated by $g_i$ and $f_i$ $\square$.

Prop VIII.2.2. In a Noetherian ring $A$ every ideal has a primary decomposition

Lemma.VIII.2.3. $A$ is Noetherian. then $\mathfrak{a} \subseteq A$ is a finite intersection of irreducible ideals

pf: Suppose not; then the set of ideals in $A$ for which the lemma is false is non-empty.

By the Noetherian property, we have a maximal element $\mathfrak{a}$

$\Rightarrow \mathfrak{a} = \mathfrak{b} \cap \mathfrak{c}$ $\Rightarrow$ $\mathfrak{b} \supseteq \mathfrak{a}$ and $\mathfrak{c} \supseteq \mathfrak{a}$ $\Rightarrow$ $\mathfrak{b}, \mathfrak{c}$ is a finite intersection of irreducible ideals, contraction $\blacksquare$.

Lemma. VIII.2.4 In a Noetherian ring $A$, every irreducible ideals is primary.

Pf: Just to show that if $0$ is irreducible. then is primary

let $xy = 0$. with $y \neq 0$. then $\mathrm{Ann}(x) \subseteq \mathrm{Ann}(x^2) \subseteq \dots$ is stationary

$\Rightarrow \mathrm{Ann}(x^n) = \mathrm{Ann}(x^{n+1}) = \dots$ $\Rightarrow$ $(x^n) \cap (y) = (0)$ (if $\exists \begin{cases} a \in (y) \\ a \in (x^n) \end{cases}$ then. $ax = 0$ $\Rightarrow$ $a := bx^n$ $bx^{n+1} = 0$ $\Rightarrow b \in \mathrm{Ann}(x^{n+1}) = \mathrm{Ann}(x^n)$ $\Rightarrow bx^n = 0 = a$) $\blacksquare$

Then by (0) is irreducible $\Rightarrow$ $(x^n) = (0) \Rightarrow x \in \mathrm{nil}(A)$ $\blacksquare$

Back to the proof, we have every ideal has a primary decomposition. $\blacksquare$

## 3 Artinian Ring

**Def VIII.3.1** (Krull dimension) A finite strictly increasing sequence $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \dots \subsetneq \mathfrak{p}_n$ the length is

Then the ~~superum~~ supremum of the lengths of all chains of prime ideals in $A$.
is called the Krull dimension of $A$

**Thm VIII.3.2** (Thm of the Artinian's ring's structure) let $A$ be an Artinian ring

Then (i) $A$ ~~only~~ has only finitely maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_n$

(ii) Jacobson radical $J = \mathfrak{m}_1 \cap \dots \cap \mathfrak{m}_n$ is nilpotent and $J = \mathfrak{m}_1 \dots \mathfrak{m}_n$

(iii) $A/J \cong A/\mathfrak{m}_1 \times \quad \times A/\mathfrak{m}_n$.

(iv) ~~let~~ $A \cong A_1 \times \dots \times A_n$. $A_i$ is a local Artinian ring. with maximal ideal $\mathfrak{m}_i A_i$ nilpotent

(v.) $A$ is a Noetherian ring.